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KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Kinematics -1D & Calculus

(English Medium)

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KINEMATICS-1D & CALCULUS

KEY CONCEPT

Kinematics

Study of motion of objects without taking into account the factor which cause the motion (i.e. nature of force).

Motion

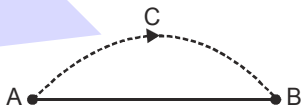
If a body changes its position with time, it is said to be moving else it is at rest. Motion is always relative to the observer.

Motion is a combined property of the object under study and the observer. There is no meaning of rest or motion without the viewer. In other words absolute motion or rest is meaningless.

DISTANCE AND DISPLACEMENT

Total length of path covered by the particle, in definite time interval is called distance.

Let a body moves from A to B via C. The length of path ACB is called the distance travelled by the body.

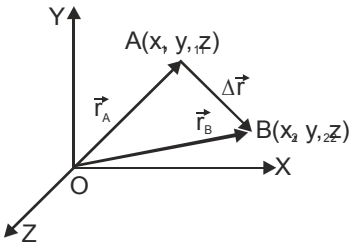


Displacement

$\vec{AB}$ is its displacement vector or displacement that is the minimum distance and directed from initial position to final position.

Displacement in terms of position vector

Let a body is displaced from $A(x_1, y_1, z_1)$ to $B(x_2, y_2, z_2)$ then its displacement is given by vector $\vec{AB}$.



From ΔOAB $\vec{r}_A + \Delta\vec{r} = \vec{r}_B \Rightarrow \Delta\vec{r} = \vec{r}_B - \vec{r}_A$

$\vec{r}_B = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$ and $\vec{r}_A = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$

$\Delta\vec{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k} \Rightarrow \Delta\vec{r} = \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k}$

Important points

- Distance is a scalar while displacement is a vector.
- Distance depends on path while displacement is independent of path but depends only on final and initial position.
- For a moving body, distance can't have zero or negative values but displacement may be +ive, –ive or zero.
- For a moving / stationary object distance can't be decreasing.
- If motion is in straight line without change in direction then
  distance = |displacement|i.e. magnitude of displacement.
- Magnitude of displacement may be equal or less than distance but never greater than distance.
  distance $\geq$ |displacement|

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SPEED :

Speed is the rate of change of distance with respect to time.

Average speed :

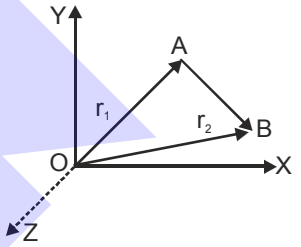
Speed is distance travelled per unit time. Average speed of a trip $v_{av} = \frac{\text{Total travelled distance}}{\text{Total time taken}}$

VELOCITY :

Velocity is the rate of change of displacement with respect to time.

Average Velocity

The average velocity of a particle in a time interval t_1 to t_2 is a defined as its displacement divided by the time interval. Let a particle is at a point A at time t_1 and B at time t_2. Position vectors of A and B are $\vec{r}_1$ and $\vec{r}_2$. The displacement in this time interval is the vector $\vec{AB} = (\vec{r}_2 - \vec{r}_1)$. The average velocity in this time interval is, $\vec{v}_{av} = \frac{\text{displacement vector}}{\text{time interval}}$

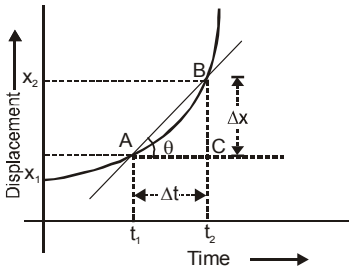


$$\vec{v}_{av} = \frac{\vec{AB}}{t_2 - t_1} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\Delta\vec{r}}{\Delta t}$$

here $\vec{AB} = \vec{OB} - \vec{OA} = \vec{r}_2 - \vec{r}_1$ = change in position vector.

Instantaneous velocity

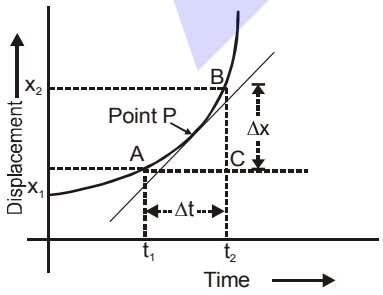
The velocity of the object at a given instant of time or at a given position during motion is called instantaneous velocity $\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$



From fig., the average velocity between points A and B is

$$v_{av} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t} = \text{slope of secant AB} = \tan\theta$$

$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} = \text{slope of tangent at P} = \tan\alpha$$



Magnitude of instantaneous velocity is the instantaneous speed.

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ACCELERATION :

Acceleration

The acceleration is rate of change of velocity or change in velocity per unit time interval.

Velocity is a vector quantity hence a change in its magnitude or in direction or in both, will change the velocity .

Uniform acceleration :

An object is said to be moving with a uniform acceleration if its velocity changes by equal amounts in equal intervals of time, irrespective of duration of intervals.

Variable acceleration :

An object is said to be moving with a variable acceleration if its velocity changes by unequal amounts in equal intervals of time, irrespective of duration of intervals..

Average Acceleration :

When an object is moving with a variable acceleration, then the average acceleration of the object for the given motion is defined as the ratio of the total change in velocity of the object during motion to the total time taken

$$\text{Average Acceleration} = \frac{\text{total change in velocity}}{\text{total time taken}}$$

Suppose the velocity of a particle is $\vec{v}_1$ at time t_1 and $\vec{v}_2$ at time t_2. Then $\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta\vec{v}}{\Delta t}$

Instantaneous Acceleration :

The acceleration of the object at a given instant of time or at a given point of motion, is called its instantaneous acceleration. Suppose the velocity of a particle at time $t_1 = t$ is $\vec{v}_1 = \vec{v}$ and becomes

$\vec{v}_2 = \vec{v} + \Delta\vec{v}$ at time $t_2 = t + \Delta t$, Then, $\vec{a}_{av} = \frac{\Delta\vec{v}}{\Delta t}$

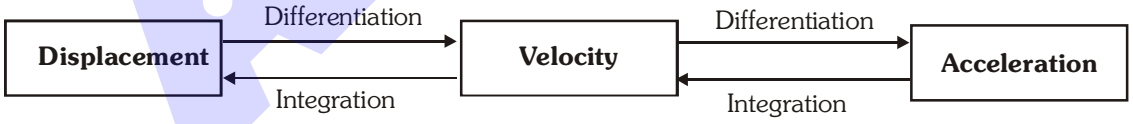
If Δt approaches to zero then the rate of change of velocity will be instantaneous acceleration.

Instantaneous acceleration $\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right) = \frac{d^2\vec{r}}{dt^2}$

Use of Mathematical Tools in Solving Problems of One–Dimensional Motion

If displacement–time equation is given, we can get velocity–time equation with the help of differentiation. Again, we can get acceleration–time equation with the help of differentiation.

If acceleration–time equation is given, we can get velocity–time equation by integration. From velocity equation, we can get displacement–time equation by integration.



Equations of motion (motion with constant acceleration)

$\vec{v} = \vec{u} + \vec{a}t$

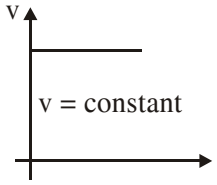
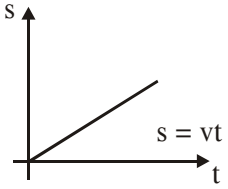
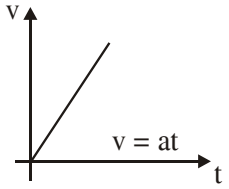
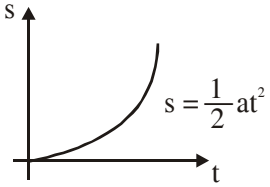
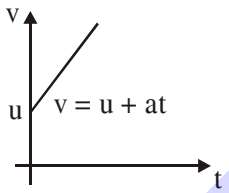
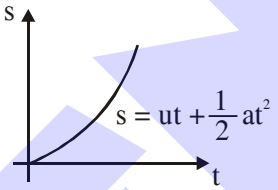
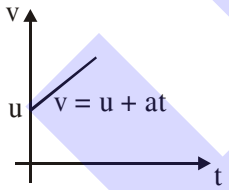
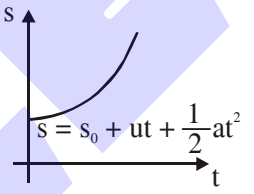
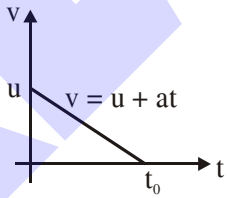
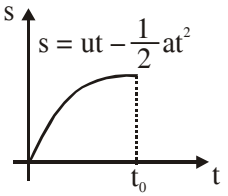
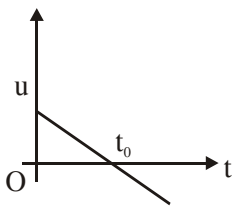
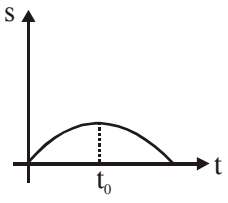
$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2$ or $\vec{s} = \vec{v}t - \frac{1}{2}\vec{a}t^2$ or $\vec{s} = \left(\frac{\vec{v} + \vec{u}}{2}\right)t \Rightarrow v^2 = u^2 + 2as$

$\boxed{v^2 = u^2 + 2\vec{a}.\vec{s}}$

$\boxed{s_{n^{th}} = u + \frac{a}{2}(2n - 1)}$

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Some important graphs

S.N.	Different Cases	v–t graph	s–t graph	Important Points
1.	Uniform motion			(i) Slope of s – t graph = v = constant (ii) In s – t graph s = 0 at t = 0
2.	Uniformly accelerated motion with u = 0 at t = 0			(i) u = 0, i.e. v = 0at t = 0 (ii) u = 0, i.e., slope of s – t graph at t = u, should be zero (iii) a or slope of v – t graph is constant
3.	Uniformly accelerated with $u \neq 0$ at t = 0			(i) $u \neq 0$, i.e., v or slope motion of s – t graph at t = 0 is not zero (ii) v or slope of s – t graph gradually goes on increasing.
4.	Uniformly accelerated motion with u ≠ 0 and $s = s_0$ at t = 0			(i) $s = s_0$ at t = 0
5.	Uniformly retarded motion till velocity becomes zero			(i) Slope of s – t graph at t = 0 gives u (ii) Slope of s – t graph at $t = t_0$ becomes zero (iii) In this case u can't be zero.
6.	Uniformly retarded then accelerated in opposite direction			(i) At time $t = t_o$, v = 0 or slope of s – t graph is zero (ii) In s – t graph slope or velocity first decreases then increases with opposite sign.

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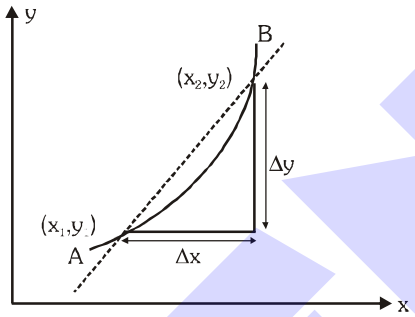
CALCULUS

Calculus is the study of how things change. In this we study the relationship between continuously varying functions.

DIFFERENTIAL CALCULUS:

The purpose of differential calculus to study the nature (i.e., increase or decrease) and the amount of variation in a quantity when another quantity (on which first quantity depends) varies independently.

Average rate of change

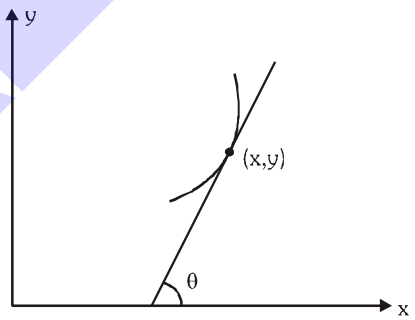


Let a function y = f(x) be plotted as shown in figure. Average rate of change in y w.r.t. x in interval $[x_1, x_2]$ is

$$\text{Average rate of change} = \frac{\text{change in y}}{\text{change in x}} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

= slope of chord AB.

Instantaneous rate of change



It is defined as the rate of change in y with x at a particular value of x. It is measured graphically by the slope of the tangent drawn to the y-x graph at the point (x,y) and algebraically by the first derivative of function y =f(x).

$$\text{Instantaneous rate of change} = \frac{dy}{dx} = \text{slope of tangent} = \tan\theta$$

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First Derivatives of Commonly used Functions

- $y = \text{constant} \Rightarrow \frac{dy}{dx} = 0$
- $y = x^n \Rightarrow \frac{dy}{dx} = nx^{n-1}$
- $y = e^x \Rightarrow \frac{dy}{dx} = e^x$
- $y = \ln x \Rightarrow \frac{dy}{dx} = \frac{1}{x}$
- $y = \sin x \Rightarrow \frac{dy}{dx} = \cos x$
- $y = \cos x \Rightarrow \frac{dy}{dx} = -\sin x$
- $y = \tan x \Rightarrow \frac{dy}{dx} = \sec^2 x$
- $y = \cot x \Rightarrow \frac{dy}{dx} = -\text{cosec}^2 x$

Method of Differentiation or Rules of Differentiation

(i) Function multiplied by a constant i.e., $y = kf(x) \Rightarrow \frac{dy}{dx} = kf'(x)$

Sum or Subtraction of Two functions

i.e., $y = f(x) \pm g(x) \Rightarrow \frac{dy}{dx} = f'(x) \pm g'(x)$

Product of two functions : Product rule

$y = f(x) \cdot g(x) \Rightarrow \frac{dy}{dx} = f(x) \cdot g'(x) + f'(x) \cdot g(x)$

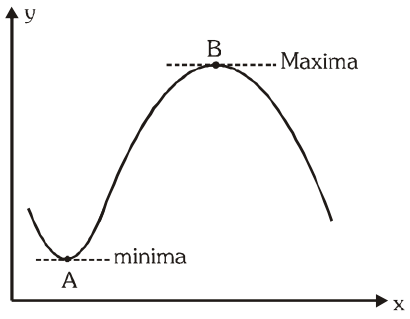
Division of Two Functions : Quotient Rule

$$y = \frac{f(x)}{g(x)} \Rightarrow \frac{dy}{dx} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Function of Functions : Chain rule

Let f be a function of x, which in turn is a function of t. The first derivative of f w.r.t. t is equal to the product of $\frac{df}{dx}$ and $\frac{dx}{dt}$ Therefore $\frac{df}{dt} = \frac{df}{dx} \times \frac{dx}{dt}$

Maximum and Minimum value of a Function



Higher order derivatives are used to find the maximum and minimum values of a function. At the points of maxima and minima, first derivative becomes zero.

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At point 'A' (minima) : As we see in figure, in the neighborhood of A, slope increases so $\frac{d^2y}{dx^2} > 0$.

Condition for minima : $\boxed{\frac{dy}{dx} = 0 \text{ and } \frac{d^2y}{dx^2} > 0}$

At point 'B' (maxima) : As we see in figure, in the neighborhood of B, slope decreases so $\frac{d^2y}{dx^2} < 0$

Condition for maxima : $\boxed{\frac{dy}{dx} = 0 \text{ and } \frac{d^2y}{dx^2} < 0}$

RATE OF CHANGE OF A VECTOR WITH TIME

It is derivative of a vector function with respect to time. Cartesian components of a time dependent vector, if given as function of time as $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$, the time rate of change can be calculated according to equation

$$\frac{d\vec{r}(t)}{dt} = \frac{dx(t)\hat{i}}{dt} + \frac{dy(t)\hat{j}}{dt} + \frac{dz(t)\hat{k}}{dt}$$

Methods of Differentiation of Vector Functions

Methods of differentiation of scalar functions are also applicable to differentiation of vector functions.

1. $\frac{d}{dt}\left(\vec{F} \pm \vec{G}\right) = \frac{d\vec{F}}{dt} \pm \frac{d\vec{G}}{dt}$
2. $\frac{d}{dt}\left(\vec{F} \cdot \vec{G}\right) = \frac{d\vec{F}}{dt} \cdot \vec{G} + \vec{F} \cdot \frac{d\vec{G}}{dt}$
3. $\frac{d}{dt}\left(X\vec{F}\right) = \frac{dX}{dt}\vec{F} + X\frac{d\vec{F}}{dt}$ Here X is a scalar function of time.

INTEGRAL CALCULUS

Integration is the reverse process of differentiation. By help of integration we can find a function whose derivative is known. Consider a function F(x) whose differentiation w.r.t. x is equal to f(x) then

$$\int f(x)\,dx = F(x) + c$$

here c is the constant of integration and this is called indefinite integration.
Few basic formulae of integration are :

$\int x^n dx = \frac{x^{n+1}}{n+1} + c$ $\qquad$ $\int \frac{1}{x} dx = \ln|x| + c$

$\int \sin x dx = -\cos x + c$ $\qquad$ $\int \cos x dx = \sin x + c$

$\int \sec^2 x dx = \tan x + c$ $\qquad$ $\int e^x dx = e^x + c$

$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c$ $\qquad$ $\int \frac{dx}{ax+b} = \frac{\ln(ax+b)}{a} + c$

$\int \sin(ax+b) dx = \frac{-\cos(ax+b)}{a} + c$ $\qquad$ $\int \cos(ax+b) dx = \frac{\sin(ax+b)}{a} + c$

$\int \sec^2(ax+b) dx = \frac{\tan(ax+b)}{a} + c$ $\qquad$ $\int e^{ax+b} dx = \frac{e^{ax+b}}{a} + c$

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Definite Integration

When a function is integrated between a lower limit and an upper limit, it is called a definite integral. Consider a function F(x) whose differentiation w.r.t. x is equal to f(x), in an interval $a \le x \le b$ then

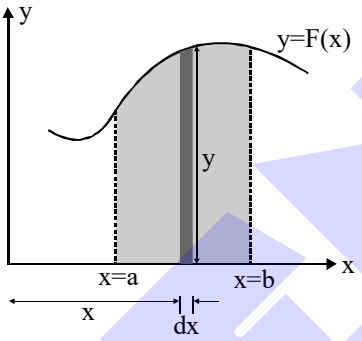
$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Area Under a Curve and Definite Integration

Area of small shown element = ydx = f(x) dx

If we sum up all areas between x=a and x= b then

$\int_a^b f(x)\,dx$ = shaded area between curve and x-axis.



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SUPPLEMENT FOR DIFFERENTIATION

BASIC DIFFERENTIATION:

1. $\frac{d}{dx}(5x^6) =$

2. $\frac{d}{dx}(3\sin x + 4\cos x)$

3. $\frac{d}{dx}(6x^4 - x^3 + 5) =$

4. $\frac{d}{dx}(4\sqrt{x} - 2x - 7) =$

5. $\frac{d}{dx}\left(5 - \frac{5}{x}\right) =$

6. $\frac{d}{dx}(e^x + \ln x)$

7. $\frac{d}{dx}\left(\frac{1}{x^2} + 4\sqrt{x^3}\right) =$

8. $\frac{d}{dx}(x - 3\sin x) =$

9. Find thwe value of $\frac{dy}{dx}$ at $x = \frac{\pi}{2}$ if $y = \frac{x^2}{\pi} + 4\cos x$.

10. Find the value of $\frac{dy}{dx}$ if $y = 2e^x + 3x$ at

(i) $x = 0$ (ii) $x = 1$

11. $y = 3x^2 + 5x - 10$. Find

(i) $\frac{dy}{dx}$ (ii) $\frac{d^2y}{dx^2}$ (iii) $\frac{d^3y}{dx^3}$

12. $y = x + 2$. Find $\frac{d}{dx}y^2$

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PRODUCT RULE:

13. $\frac{d}{dx}(x^2 \sin x) =$

14. $\frac{d}{dx}(e^2 \cos x) =$

15. If $y = (x^2 + 3x - 4)\sin x$, find $\frac{dy}{dx}$

16. $\frac{d}{dx}(x^2 \ln x) =$

17. $\frac{d}{dx}(\sin x \cos x) =$

18. $\frac{d}{dx}\left[(x^2 + 2x)e^x\right] =$

19. $(\cos x)\frac{d}{dx}(x \tan x) =$

20. $\frac{d}{dx}\left[\sqrt{x}(x^2 + 5)\right] =$

21. $\frac{d}{dx}(\sin^2 x) =$

QUOTIENT RULE:

22. $\frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) =$

23. $\frac{d}{dx}\left(\frac{x + x^2}{\sqrt{x}}\right) =$

24. $\frac{d}{dx}\left(\frac{1}{\cos x}\right)$

25. $\frac{d}{dx}\left(\frac{\cos x}{\cos x - \sin x}\right)$

26. $\frac{d}{dx}\left(\frac{e^x}{\sin x}\right) =$

27. $\frac{d}{dx}\left(\frac{e^x}{x^2 + 1}\right) =$

28. $y = \left(\frac{x^2 + 1}{x^2 - 1}\right)$ find $\frac{dy}{dx}$ at $x = 2$.

29. If $y = \frac{1 + \sin x}{1 - \sin x}$ find $\frac{dy}{dx}$ at $x = \frac{\pi}{6}$

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CHAIN RULE:

30. $\frac{d}{dx}(2x+3)^4 =$

31. $\frac{d}{dx}(\sin^3 x) =$

32. $\frac{d}{dx}e^{4x+3} =$

33. $\frac{d}{dx}(x^2+2)^3 =$

34. $\frac{d}{dx}\ln(3x^2+x)$

35. $\frac{d}{dx}\sin^2(4x+5)$

36. If $v = \frac{dx}{dt}$, $v = \sin x$, find $\frac{dv}{dt}$

37. $y = \frac{2x}{\sqrt{x^2+3}}$ find $\frac{dy}{dx}$ at $x = 1$.

38. $\frac{d}{dx} = \ln(\sin x)$

39. $y = \ln(\sec x + \tan x)$ then find $\frac{dy}{dx}$

MAXIMA AND MINIMA:

40. $y = 4 - 2x + x^2$, find the minimum value of y.

41. $y = 3 + 4x - x^2$, find the maxima value of y

42. $y = 3\sin x + 4\cos x$ find the maxima value of y.

43. $y = \frac{2\sqrt{5}}{\cos x + 3\sin x}$, find the minimum value of y.

44. If $y = \frac{x^3}{3} - 4x$, find y_{max} and y_{min}.

45. If x and y both are positive $y = x + \frac{1}{x}$, then which of the following is possible:

(A) $y = 1$ (B) $y = \frac{1}{2}$ (C) $y = \frac{3}{2}$ (D) $y = 3$

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SUPPLEMENT FOR INTEGRATION

BASIC INTEGRATION:

1. $\int (3\sqrt{x}+2)dx =$

2. $\int (3x^2+2x+3)dx =$

3. $\int \sqrt{z}\left(z^2-\frac{1}{4z}\right)dz =$

4. $\int \left(4e^y+15-\frac{1}{6y}\right)dy$

5. $\int \left(\sqrt{x}+\frac{1}{3\sqrt{x}}\right)dy$

6. $\int \left(\frac{8}{x}-\frac{5}{x^2}+\frac{6}{x^3}\right)dx =$

7. $\int \left(\frac{x^4+4}{x^2}\right)dx =$

8. $\int \frac{1}{\cos^2\theta}d\theta =$

9. $\int 2\cos^2\left(\frac{\theta}{2}\right)d\theta =$

10. $\int 2\sin^2\left(\frac{\theta}{2}\right)d\theta$

INTEGRATION USING SUBSTITUTION:

11. $\int 9\sin(3x)dx$

12. $\int 12\cos(4\theta)d\theta$

13. $\int 4e^{-7x}dx =$

14. $\int -5\cos(\pi x)dx$

15. $\int (3x^2+4e^{2x+3})dx$

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16. $\int \frac{4}{4x+3}dx$

17. $\int\left(\frac{1}{4x+3}+\frac{1}{4x+5}\right)dx$

18. $\int[\cos(5x-3)+\cos(2x+5)]dx$

19. $\int \frac{3}{x^2+3}dx$

20. $\int \frac{2x}{(4x+3x^2)^2}dx$

21. $\int \tan x\,dx$

(Hint: Write $\tan x=\frac{\sin x}{\cos x}$)

22. $\int \sec x dx$

23. $\int (\sin^4 x)\cos x dx$

24. $\int \cos^3 x dx$

25. $\int \sin^3 x dx$

DEFINITE INTEGRATION:

26. If $\theta_1=0°$ and $\theta_2=30°$, $\int_{\theta_1}^{\theta_2}(2+\cos\theta)d\theta$

27. $\int_1^4(5x^2-8x+5)dx$

28. $\int_1^9(x\sqrt{x}+2x+3)dx$

29. $\int_{-1}^{2}(1+3t)t^2dt$

30. $\int_0^4 e^{x/2}dx$

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31. $\int_0^{\pi/4} \cos\left(3x+\frac{\pi}{4}\right)dx$

32. $\int_0^{3} \frac{4}{2x+3}dx$

33. $\int_0^{\ln 2} 2e^{2x}dx$

34. $\int_0^{1} \frac{9}{(2x+1)^2}dx$

35. $\int_{\pi/6}^{\pi/3} \sin\left(4x+\frac{\pi}{6}\right)dx$

36. $\int_e^{3} \frac{1}{x\ln x}dx$

37. If $y = 2x\sin 2x + \cos 2x$,

(a) Find $\frac{dy}{dx}$

(b) using the result of part (a) calculate $\int_0^{\pi/4} (x\cos 2x)dx$ =

38. If $\frac{8x}{4x-3} = A + \frac{6}{4x-3}$

(a) Find the value of A. (b) Calculate $\int_1^{3} \frac{8x}{4x-3}dx$

{ln3 = 1.1}

39. $\int_1^{2} \frac{dx}{\sqrt{4x+1}}$

40. $\int_1^{2} \frac{dx}{(4+x^2)^{3/2}}$

(Hint: use substitution $x = a\tan\theta$)

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